Roll No.

Set-1

END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech Petroleum Engineering *Semester: IV*

Name of the Paper: Mud Logging *Course Code:* TPE-413

Time: Three Hours MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1		
a.	How drilling fluid with the proper characteristics can support the formation walls during drilling? How many type of mud is used in drilling? Explain the five properties of the drilling mud.	CO-1
b.	What is Lag time and Lag strokes? Why it is important in mud logging? How to calculate it?	CO-2
c.	Explain the principle of sensor. What are the various sensors used for recording mud logging data? What is the function of the sensors in mud logging operations?	CO-3
Q2		
a.	What are the various mud logs recorded during drilling? Explain with a diagram the various components of mud log to be put in various tracks for interpretation?	CO-3
b.	How many types of pressure exist in the formation? Draw pressure gradient plot of all formation pressure? What is the various reason for over pressure?	CO-3
c.	What is Coring? Why it is required? What are the primary methods for wireline coring? which method is more suitable in wireline coring?	CO-4
Q3		
a.	What is the various information we get from the cutting during drilling operations? What are wet cut and dry cut sample? Explain sample lag estimation in details.	CO-4
b.	What is pore pressure? Explain the various techniques used in mud logging for pore pressure prediction during drilling a well. What is drilling exponent or d-exponent? Write its formula for calculation.	CO-4
c.	What is conventional coring? Write the core handling process in detail. What is the precaution required for core handing?	CO-4

Q4		
a.	Write the fundamental gas properties in detail. What is the difference between conventional gas reservoirs and CBM gas reservoir?	CO-4
b.	What are gas hydrates? How are they formed? What is the essential condition for gas hydrate?	CO-5
c.	What is Hydrostatic pressure, Pore pressure, abnormal pressure. Over pressure, Under pressure in details. How to calculate formation pressure using d-exponent?	CO-5
Q5		
a.	Why water changes its flow from laminar to turbulent when is heated? Explain Viscosity, Reynolds Number in details.	CO-5
b.	How many types of fluid we have? What is the role of drilling fluid during drilling operation? What is the condition for using different drilling fluids in drilling process?	CO-5
c.	What are the various sample quality and examination techniques available? Write the sample analysis procedure in detail.	CO-5